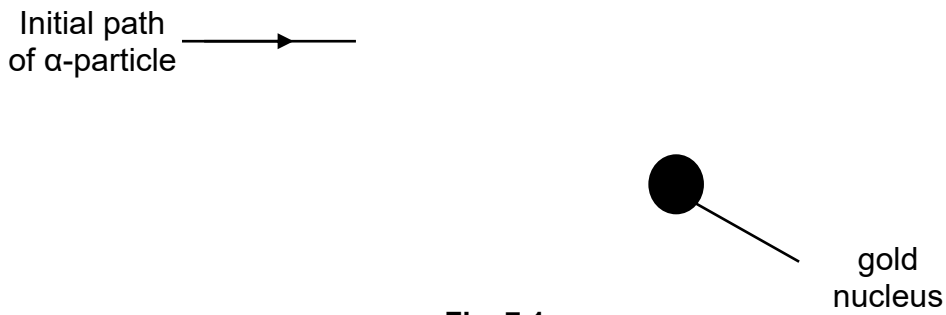
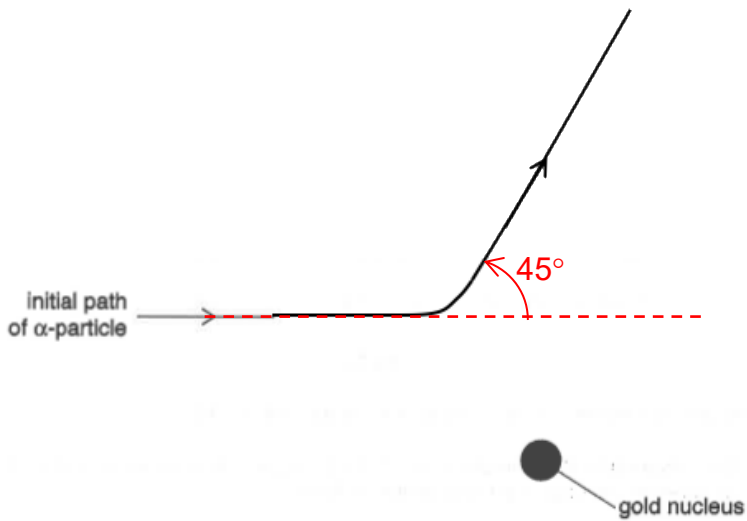


7	(a)	In the Rutherford α -particle scattering experiment, α -particles are emitted from a source and travel towards a thin gold foil.
	(i)	An α-particle is deflected through an angle of approximately 45° as it passes near a stationary gold nucleus. On Fig. 7.1, sketch the path of the α-particle as it passes the gold nucleus.  The diagram shows a horizontal arrow pointing to the right, labeled 'Initial path of α-particle'. To the right of this arrow is a solid black circle labeled 'gold nucleus'. Fig. 7.1
		Solution:  The diagram shows the same setup as Fig. 7.1, but with a curved arrow representing the path of the α-particle. The path starts horizontally from the left, then curves upwards and to the right, ending at an angle. A red dashed line extends horizontally from the start of the curve. A red arc indicates the angle between the initial horizontal path and the final path, labeled 45°. Marking points: (B1 - both have to be present) - Deflect away from the nucleus, since like charges repel. - Deviation through 45°.
	(ii)	Only a small proportion of the α-particles incident on the metal foil are deflected through large angle deflections greater than 90°. Explain the following phenomenon.
		

			[2]
		Solution: The atom contains a small, positively charged nucleus that is very small compared to the overall size of the atom, so only a small proportion of alpha particles come close enough to it. These alpha particles experience strong electrostatic repulsion, which can cause deflections greater than 90° since alpha particles are also positively charged.	B1 B1
		(ii)	In the α-particle scattering experiment, a large number of alpha particles are directed at the metal foil. Explain why a large number of alpha particles is necessary.
			[2]
		Solution: Large angle deflections are rare / occur for only a small proportion of alpha particles. A large number ensures that enough deflections are observed to draw reliable conclusions / obtain statistically significant results.	B1 B1
		(b)	An α-particle with kinetic energy $7.7 \times 10^{-13} \text{ J}$ is directed at a stationary gold nucleus ($^{197}_{79}\text{Au}$). Determine the minimum separation possible between this α-particle and the gold nucleus.
			separation = m [3]
		Solution: Given: • Charge of α-particle: $Q_1 = 2e = 2 \times 1.60 \times 10^{-19} \text{ C}$ • Charge of gold nucleus: $Q_2 = 79e = 79 \times 1.60 \times 10^{-19} \text{ C}$	

		At the start of the alpha particle's motion, when it is far away from the nucleus, the electric potential energy is taken to be zero. By conservation of energy, Gain in EPE = loss in KE $\frac{Q_1 Q_2}{4\pi\epsilon_0 r} = E_k$ $r = \frac{Q_1 Q_2}{4\pi\epsilon_0 E_k} = \frac{(2 \times 1.60 \times 10^{-19}) \times (79 \times 1.60 \times 10^{-19})}{4\pi(8.85 \times 10^{-12}) 7.7 \times 10^{-13}}$ [correct substitution] $r = 4.7 \times 10^{-14} \text{ m}$	B1 M1 A1
	(c)	The metal foil is changed from gold ($^{197}_{79}\text{Au}$) to carbon ($^{12}_6\text{C}$), while the α-particle energy is kept the same. State and explain how the number of large-angle deflections would change.	
		[2]	
		Solution: As carbon nuclei have a smaller positive charge (atomic number 6) compared to gold (atomic number 79), the electrostatic repulsion between the α-particle and the carbon nucleus is weaker. As a result, fewer α-particles are deflected through large angles, so the number of large-angle deflections decreases.	M1 A1

[Total: 10]

22

Section B

Answer **one** question from this Section in the spaces provided.
